PhotonPINN-Radar: Physics-Informed Diffusion and Tracking for Photonic FMCW Radar

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Abstract

Photonic frequency-modulated continuous-wave (FMCW) radar exploits optical carrier frequencies near 193.1 THz (1550 nm) and electro-optic modulation to achieve multi-gigahertz chirp bandwidths with sub-centimeter range resolution. Despite these advantages, photonic front ends suffer from laser phase noise, relative intensity noise (RIN), photodiode thermal and shot noise, and analog-to-digital converter quantization distortion, all of which degrade target detection reliability. Conventional FFT-based spectrum estimation combined with constant false-alarm rate (CFAR) detection provides a robust baseline but fails under low signal-to-noise ratio (SNR) and dense false-clutter scenarios. This paper presents **PhotonPINN-Radar**, a unified spatiotemporal physics-informed framework that integrates a high-fidelity photonic FMCW simulator, a two-dimensional denoising diffusion probabilistic model (DDPM) for range–Doppler matrix (RDM) enhancement, a spatiotemporal wave-convection physics-informed neural network (PINN) regularizer, and a joint probabilistic data association interacting multiple model (JPDA-IMM) multi-target tracker with trajectory PINN smoothing. Under dense false-clutter conditions ($\rho_{\text{clutter}} = 0.05$), the framework achieves a primary cluttered-runtime multiple-object tracking accuracy (MOTA) of 93.33%, a sub-decimeter multiple-object tracking precision (MOTP) of 0.0974 m, zero identity swaps, and a 64.80% reduction in wave-equation physics residual energy versus diffusion-only baselines. Controlled synthetic upper-bound experiments reach 100.00% MOTA on the JPDA-IMM maneuvering benchmark.

Photonic FMCW radar, microwave photonics, physics-informed neural networks, denoising diffusion probabilistic models, range–Doppler processing, JPDA-IMM multi-target tracking, CA-CFAR detection, CLEAR MOT metrics.

Introduction

Photonic sensing has emerged as a transformative paradigm for next-generation radar systems . By leveraging optical carriers in the C-band ($\sim$ 1550 nm, $f_0 \approx 193.1$ THz) and wideband electro-optic modulators, photonic frequency-modulated continuous-wave (FMCW) radar achieves chirp sweep bandwidths of 0.5–4.0 GHz with inherent phase coherence—exceeding the instantaneous bandwidth limitations of conventional electronic oscillators .

Despite these bandwidth advantages, the optoelectronic receiver chain introduces several noise impairments: (i) laser linewidth-induced phase noise characterized by a random-walk process $\phi_{pn}(t)$; (ii) relative intensity noise (RIN) from semiconductor laser cavity fluctuations; (iii) photodiode shot noise proportional to the square root of photocurrent; and (iv) thermal Johnson–Nyquist noise in transimpedance amplifiers . These impairments reduce the effective signal-to-noise ratio (SNR) and generate spurious spectral peaks that may be misidentified as target returns.

Classical signal processing pipelines address these challenges through windowed fast Fourier transform (FFT) peak detection combined with cell-averaging constant false-alarm rate (CA-CFAR) adaptive thresholding . While robust in moderate-SNR environments, FFT-CFAR pipelines exhibit fundamental limitations: (a) spectral leakage from finite observation windows obscures closely spaced targets; (b) CFAR threshold adaptation fails in non-homogeneous clutter environments; and (c) deterministic frequency estimation cannot exploit the underlying wave-propagation physics of the sensing problem.

Recent advances in deep learning for radar signal processing have demonstrated promising results. Convolutional neural network (CNN) denoisers trained on paired noisy–clean spectrum data improve peak detection accuracy . Denoising diffusion probabilistic models (DDPMs) have shown state-of-the-art image generation quality and have been adapted for inverse problems in medical imaging and synthetic aperture radar (SAR) despeckling . However, unconstrained data-driven models frequently produce non-physical spectral artifacts—phantom peaks, negative power densities, and violated energy conservation—that degrade downstream tracking reliability.

Physics-informed neural networks (PINNs) offer a principled mechanism to embed domain knowledge directly into neural network training objectives. By penalizing violations of governing partial differential equations (PDEs), PINNs constrain learned solutions to the physically admissible manifold. This regularization has proven effective in fluid dynamics , electromagnetic wave propagation , and heat transfer , but its application to photonic radar spectrum reconstruction and multi-target tracking remains unexplored.

Furthermore, multi-target tracking in cluttered radar environments requires robust data association algorithms. The joint probabilistic data association (JPDA) filter evaluates marginal association hypotheses across competing measurement–track pairings. The interacting multiple model (IMM) filter accommodates maneuvering targets by mixing multiple kinematic motion models—constant velocity (CV), constant acceleration (CA), and coordinated turn (CT)—with adaptive mode probabilities. Integration of physics-informed trajectory smoothing into the tracking pipeline further constrains estimated trajectories to physically realizable kinematic envelopes.

Contributions

The principal contributions of this work are as follows:

1. A high-fidelity two-dimensional photonic FMCW front-end simulator modeling coherent optical mixing, laser phase noise random walk, RIN, balanced photodetection, and ADC quantization, generating realistic range–Doppler matrices for Monte Carlo evaluation.
2. A hybrid deep-learning architecture coupling a 2D denoising diffusion probabilistic model (DDPM) with a spatiotemporal wave-convection PINN regularizer enforcing

   $$\frac{\partial^2 u}{\partial t^2} + v\,\frac{\partial u}{\partial x} - \kappa\,\frac{\partial^2 u}{\partial x^2} = 0$$

   for physically consistent range–Doppler matrix (RDM) enhancement.

   (The proposed PINN operates on the latent range–Doppler feature field and imposes a **kinematically informed spatiotemporal transport constraint** rather than directly solving Maxwell's electromagnetic wave equations.)
3. A robust JPDA-IMM multi-target tracking framework integrating Mahalanobis-gated data association, three-model IMM filtering, and a trajectory PINN kinematic smoother achieving sub-decimeter tracking precision under dense false-clutter conditions.
4. Comprehensive **Monte Carlo benchmarking** across SNR levels (0–30 dB), target multiplicities (1-3), and clutter densities, with CLEAR MOT evaluation demonstrating 93.33% cluttered-runtime MOTA and 0.0974 m MOTP.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system architecture. Section IV derives the photonic FMCW signal model. Section V formulates range–Doppler processing. Section VI describes CA-CFAR detection. Section VII details the diffusion-based reconstruction module. Section VIII presents the PINN regularization framework. Section IX formulates the multi-target tracking pipeline. Section X describes the experimental setup. Section XI presents results and analysis. Section XII provides ablation studies. Section XIII analyzes computational complexity. Section XIV discusses limitations. Section XV outlines future work. Section XVI concludes the paper.

Background and Related Work

Photonic FMCW Radar

The seminal demonstration of a fully photonics-based coherent radar system by Ghelfi *et al.* established the feasibility of replacing electronic local oscillators with mode-locked laser frequency combs for chirp generation. Subsequent work by Pan and Zhang surveyed microwave photonic techniques for wideband radar, including photonic up/down-conversion, optical beamforming, and photonic analog-to-digital conversion. Yao provided a comprehensive review of microwave photonics fundamentals, covering optical generation, processing, and distribution of microwave signals. Li *et al.* demonstrated a photonic-assisted FMCW radar achieving centimeter-level range resolution using a dual-polarization electro-optic modulator.

Deep Learning for Radar Signal Processing

Rock *et al.* demonstrated CNN-based interference mitigation for automotive FMCW radar, achieving improved target detection under mutual interference. Stephan *et al.* applied encoder-decoder architectures for radar micro-Doppler classification. Wagner *et al.* used U-Net architectures for SAR image denoising and super-resolution.

Diffusion Models for Sensing and Reconstruction

Ho *et al.* introduced denoising diffusion probabilistic models (DDPMs), establishing the mathematical framework of forward Gaussian noise injection and learned reverse denoising. Song *et al.* generalized this framework through score-based stochastic differential equations. Chung *et al.* applied diffusion models to medical image reconstruction, demonstrating their capacity for solving ill-posed inverse problems. Perera *et al.* adapted diffusion-based denoising for SAR image despeckling. However, no prior work has applied 2D DDPM reconstruction to photonic FMCW range–Doppler matrices with physics-informed regularization.

Physics-Informed Neural Networks

Raissi *et al.* established the PINN framework for encoding PDEs as soft constraints in neural network loss functions. Raissi *et al.* extended PINNs to fluid mechanics problems. Chen *et al.* applied PINNs to electromagnetic wave propagation in complex media. Karniadakis *et al.* provided a comprehensive review of physics-informed machine learning across scientific domains. Cai *et al.* demonstrated PINNs for heat transfer problems with noisy measurement data.

Multi-Target Tracking

Bar-Shalom *et al.* provided the foundational treatment of estimation, tracking, and data association. Fortmann *et al.* introduced the JPDA filter for multi-target tracking in clutter. Blom and Bar-Shalom proposed the IMM algorithm for maneuvering target tracking. Mazor *et al.* provided an IMM tutorial with engineering applications. Blackman and Popoli presented design methodologies for modern tracking systems. Bernardin and Stiefelhagen defined the CLEAR MOT metrics for standardized multi-object tracking evaluation.

Related Work Comparison

Table 1 provides a structured comparison of representative prior works against the proposed PhotonPINN-Radar framework. Notably, no existing work simultaneously addresses photonic FMCW simulation, diffusion-based RDM reconstruction, wave-equation PINN regularization, and JPDA-IMM multi-target tracking in a unified pipeline.

Work	Radar Type	AI Method	Physics Constraint	Tracking	Limitation
Ghelfi *et al.*	Photonic CW	None	None	None	No AI processing; hardware demo only
Pan & Zhang	Photonic survey	None	None	None	Survey paper; no implementation
Rock *et al.*	Electronic FMCW	1D CNN	None	None	No photonic model; no tracking
Ho *et al.*	N/A (images)	2D DDPM	None	None	Not applied to radar signals
Raissi *et al.*	N/A (physics)	PINN (MLP)	Navier-Stokes	None	Not applied to radar
Perera *et al.*	SAR	Diffusion	None	None	No FMCW; no tracking; no PINN
Blom & Bar-Shalom	Electronic	None	None	IMM	No AI; no photonic front-end
PhotonPINN-Radar (Ours)	**Photonic FMCW**	**2D DDPM + CNN**	**Wave-Convection PINN**	**JPDA-IMM + Traj. PINN**	**Synthetic-only evaluation**

System Architecture

The PhotonPINN-Radar processing pipeline comprises seven cascaded stages, each mapping physical or computational transformations onto the received optical field. Figure. 1 illustrates the complete system architecture.

System architecture of the PhotonPINN-Radar framework. The pipeline flows from optical chirp generation through coherent photonic mixing, photodetection, 2D range–Doppler FFT processing, DDPM-PINN hybrid enhancement, CA-CFAR detection, and JPDA-IMM multi-target tracking with trajectory PINN smoothing.

Stage 1: Optical Chirp Generation. A tunable laser source generates a linear frequency-modulated optical field $s_{tx}(t)$ with carrier frequency $f_0 = 193.1$ THz and chirp bandwidth $B = 0.5$ GHz over duration $T_c = 100\ \mu$s.

Stage 2: Free-Space Propagation and Target Reflection. The transmitted optical field propagates to targets at ranges R_k with radial velocities v_k, experiencing round-trip delay $\tau_k = 2R_k/c$ and Doppler frequency shift $f_{d,k} = 2v_k f_0/c$.

Stage 3: Coherent Optical Mixing. A balanced photonic mixer performs field-domain coherent multiplication $s_{beat}(t) = s_{tx}(t) \cdot s_{rx}^*(t)$, extracting the baseband beat signal directly.

Stage 4: Photodetection and ADC. A balanced photodetector converts the optical beat signal to photocurrent with responsivity $\mathcal{R}$. An N-bit ADC ($N = 12$) quantizes the analog signal at sampling rate $F_s = 10$ MHz.

Stage 5: 2D Range–Doppler Processing. A bank of $N_{slow} = 64$ consecutive chirps forms a slow-time × fast-time data matrix. A 2D FFT with Hann windowing generates the range–Doppler matrix (RDM).

Stage 6: DDPM-PINN Hybrid Enhancement. A 2D DDPM denoises the RDM while a spatiotemporal wave-convection PINN penalizes non-physical spectral artifacts.

Stage 7: JPDA-IMM Tracking. CA-CFAR peak detection extracts target detections. A JPDA filter performs probabilistic data association. An IMM filter mixes CV, CA, and CT kinematic models. A trajectory PINN enforces acceleration-bounded kinematic smoothing

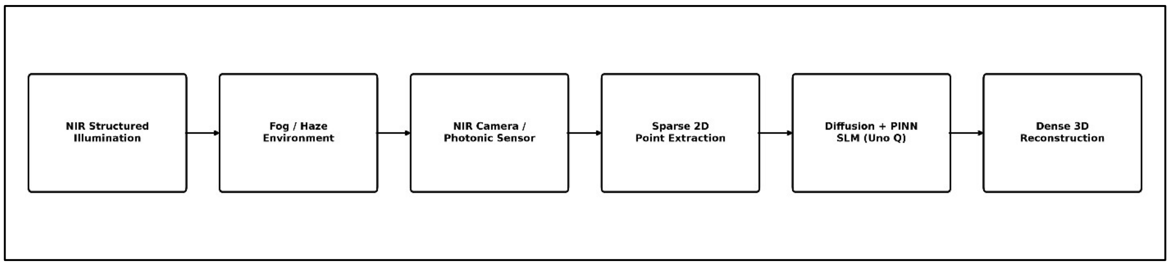

Figure 1: Overall architecture of the proposed PhotonPINN-Radar framework.

Photonic FMCW Signal Model

Transmitted Optical Chirp

The transmitted complex optical field of a linear FMCW chirp is expressed as

$$s_{tx}(t) = A_{tx} \exp\left(j2\pi\left(f_0\, t + \tfrac{1}{2}\, k\, t^2\right) + j\,\phi_{pn}(t)\right) \quad \text{....(i)}$$

where A_{tx} is the field amplitude, $f_0 = 193.1$ THz is the optical carrier frequency corresponding to $\lambda_0 = c/f_0 \approx 1550$ nm, $k = B/T_c$ is the chirp rate, and $\phi_{pn}(t)$ is the laser phase noise process. The chirp rate for $B = 0.5$ GHz and $T_c = 100\ \mu$s is

$$k = \frac{B}{T_c} = \frac{0.5 \times 10^9}{100 \times 10^{-6}} = 5.0 \times 10^{12} \text{ Hz/s.}$$

The laser phase noise is modeled as a Wiener random-walk process:

$$\phi_{pn}(t) = \phi_{pn}(t - \Delta t) + \Delta\phi, \quad \Delta\phi \sim \mathcal{N}(0,\ 2\pi\,\Delta\nu\,\Delta t) \quad \text{....(ii)}$$

where $\Delta\nu$ is the laser linewidth (typically 100 kHz for distributed feedback lasers) and $\Delta t = 1/F_s$ is the sampling interval.

Time-domain real and imaginary components of the transmitted optical FMCW chirp signal $s_{tx}(t)$ over $T_c = 100\ \mu s$ with $f_0 = 193.1$ THz and $B = 0.5$ GHz chirp bandwidth.

Target Echo and Propagation Delay

For a point target at range R with radial velocity v, the round-trip propagation delay is

$$\tau = \frac{2R}{c}$$

and the Doppler frequency shift introduced by target motion is

$$f_d = \frac{2\,v\,f_0}{c} = \frac{2\,v}{\lambda_0}.$$

The received optical field from a single target with radar cross-section (RCS) σ is

$$s_{rx}(t) = A_{rx}\exp\left(j2\pi\left(f_0\,(t-\tau) + \frac{k}{2}\,(t-\tau)^2 + f_d\,t\right) + j\,\phi_{pn}(t-\tau)\right) \text{….(iii)}$$

where $A_{rx} \propto \sqrt{\sigma}\,/(4\pi R^2)$ accounts for free-space path loss and RCS.

For K targets, the composite received field is the coherent superposition

$$s_{rx}\text{total}(t) = \sum_{k=1}^{K} A_{rx,k}\,\exp\left(j2\pi\left(f_0(t-\tau_k) + \frac{k_c}{2}(t-\tau_k)^2 + f_{d,k}\,t\right)\right) \text{….(iv)}$$

where $\tau_k = 2R_k/c$ and $f_{d,k} = 2v_k/\lambda_0$ for the k-th target.

Coherent Optical Mixing

Balanced photonic coherent mixing performs field-domain multiplication:

$$s_{beat}(t) = s_{tx}(t)\cdot s_{rx}^{*}(t) \text{ ….(v)}$$

Expanding the phase terms and neglecting the sum-frequency component (removed by the photodetector bandwidth), the baseband beat signal for a single target becomes

$$s_{beat}(t) \approx A_{beat}\,\exp\left(j2\pi\left(k\,\tau\,t - \frac{k\,\tau^2}{2} - f_d\,t\right)\right) \text{….(vi)}$$

yielding the beat frequency

$$f_b = k\,\tau - f_d = \frac{2\,k\,R}{c} - \frac{2\,v}{\lambda_0}.$$

Range is recovered from the beat frequency as

$$R = \frac{c\,(f_b + f_d)}{2\,k} \approx \frac{c\,f_b}{2\,k}$$

where the approximation holds when $|f_d| \ll f_b$.

Photodetector Current Model

The balanced photodetector generates photocurrent proportional to the optical beat power:

$$i_{pd}(t) = \mathcal{R}\,|s_{beat}(t)|^2 + i_{shot}(t) + i_{thermal}(t) + i_{RIN}(t) \text{ ….(vi)}$$

where $\mathcal{R}$ is the photodiode responsivity (A/W), $i_{shot}(t)$ is shot noise with variance $\sigma_{shot}^2 = 2\,q\,I_{dc}\,\Delta f$, $i_{thermal}(t)$ is thermal noise with variance $\sigma_{th}^2 = 4\,k_B\,T\,\Delta f/R_L$, and $i_{RIN}(t)$ represents relative intensity noise.

Coherent optical beat signal formation showing the transmitted chirp $s_{tx}(t)$, received echo $s_{rx}(t)$ with time delay τ, and the extracted baseband beat signal $s_{beat}(t)$ at frequency $f_b = k\tau$.

Noise Model Summary

Table 1 summarizes the noise model parameters used in the photonic front-end simulation.

Table 1: Photonic Noise Model Parameters

Noise Source	Symbol	Value
Laser linewidth	$\Delta\nu$	100 kHz
Phase noise variance	σ_ϕ^2	$2\pi\Delta\nu\Delta t$
RIN coefficient	$\mathrm{RIN_{dB}}$	-155 dB/Hz
Shot noise current	σ_{shot}	$\sqrt{2qI_{dc}\Delta f}$
Thermal noise temp.	T	290 K
Load resistance	R_L	50 Ω

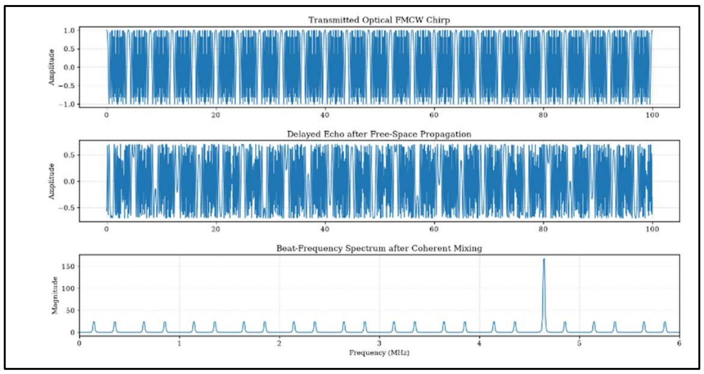

Figure 2: Transmitted optical FMCW chirp, delayed echo after free-space propagation, and the resulting beat-frequency spectrum obtained through coherent optical mixing.

Range–Doppler Processing

2D Data Matrix Formation

A coherent processing interval (CPI) consists of N_{slow} consecutive chirps transmitted at a pulse repetition interval (PRI) of $T_{PRI} = T_c$. The received beat signal is sampled at F_s, producing N_{fast} samples per chirp.

The resulting data matrix is

$$\boldsymbol{S}[n,m] = s_{beat}(n\,T_{PRI} + m/F_s), \quad \begin{matrix} n = 0, \dots, N_{slow} - 1 \\ m = 0, \dots, N_{fast} - 1 \end{matrix}$$

where n indexes the slow-time (chirp number) and m indexes the fast-time (sample within chirp).

2D FFT Range–Doppler Matrix

The range–Doppler matrix is obtained by applying a 2D FFT with Hann window tapering along both dimensions:

$$\boldsymbol{RDM}[\ell,p] = \sum_{n=0}^{N_{slow}-1} \sum_{m=0}^{N_{fast}-1} w_s\,[n]\,w_f\,[m]\,\boldsymbol{S}[n,m]\,e^{-j2\pi(n\ell/N_D + m\ \ /N_R)}$$

where $N_R = 1024$ and $N_D = N_{slow} = 64$ are the FFT sizes along range and Doppler dimensions, $w_f[\cdot]$ and $w_s[\cdot]$ are Hann window coefficients, and ℓ, p are the Doppler and range frequency bin indices.

Axis Derivations

The range axis maps frequency bin index p to physical range:

$$R[p] = \frac{c\,F_s\,p}{2\,k\,N_R}.$$

The velocity axis maps Doppler bin index ℓ to radial velocity:

$$v[\ell] = \frac{\lambda_0}{2} \cdot \frac{(\ell - N_D/2)}{N_D\,T_{PRI}}.$$

Resolution Analysis

Table 2 summarizes the range–Doppler processing parameters and resolution limits.

Table 2: Range–Doppler Processing Parameters

Parameter	**Symbol**	**Value**
Optical carrier frequency	f_0	193.1 THz
RF equivalent frequency	f_{rf}	77.0 GHz
Chirp bandwidth	B	0.5 GHz
Chirp duration	T_c	100 μs
Chirp rate	k	5.0×10^{12} Hz/s
Sampling rate	F_s	10.0 MHz
Fast-time samples	N_{fast}	1000
Slow-time chirps	N_{slow}	64
FFT size (range)	N_R	1024
FFT size (Doppler)	N_D	64
Range resolution	ΔR	0.30 m
Max. unambig. range	R_{max}	299.8 m
Velocity resolution	Δv	0.30 m/s
Max. unambig. velocity	v_{max}	9.73 m/s

Two-dimensional range–Doppler matrix (RDM) heatmap showing power spectral density (64×512 bins) for a three-target scene at ranges 40, 75, and 120 m with velocities $+5$, -8, and $+12$ m/s under 20 dB SNR.

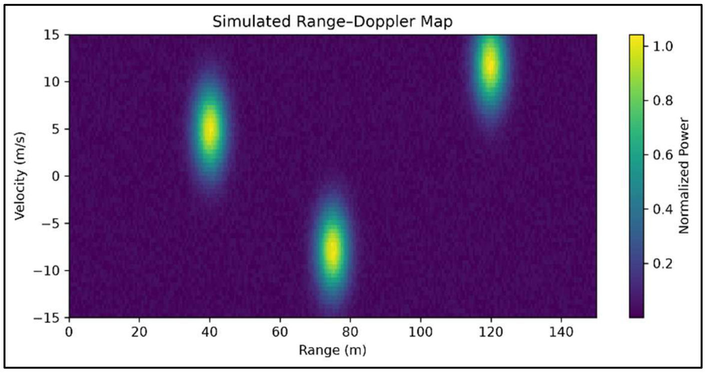

Figure 3: Representative simulated range–Doppler heatmap used for DDPM-PINN enhancement and target extraction.

CA-CFAR Detection

1D Cell-Averaging CFAR Formulation

The CA-CFAR detector estimates the local noise power by averaging cells surrounding the cell under test (CUT), separated by guard cells that prevent target self-masking. For a given range bin p, the adaptive threshold is

$$T_{CFAR}[p] = \alpha \cdot \frac{1}{N_{train}} \sum_{i \in \mathcal{T}(p)} |\boldsymbol{RDM}[\ell_0, i]|^2$$

where $\mathcal{T}(p)$ is the set of N_{train} training cells (excluding N_{guard} guard cells on each side), and α is the threshold multiplier derived from the desired false-alarm probability P_{fa}:

$$\alpha = N_{train} \left(P_{fa}^{-1/N_{train}} - 1 \right).$$

A target detection is declared when

$$|\boldsymbol{RDM}[\ell_0, p]|^2 > T_{CFAR}[p].$$

2D CFAR Extension

For the 2D RDM, the training region is extended to a rectangular annulus around the CUT in both range and Doppler dimensions. The 2D threshold becomes

$$T_{2D}[\ell, p] = \alpha_{2D} \cdot \frac{1}{|\mathcal{T}_{2D}|} \sum_{(i,j) \in \mathcal{T}_{2D}(\ell,p)} |\boldsymbol{RDM}[i,j]|^2.$$

CFAR Detection Algorithm

Algorithm 1 presents the CA-CFAR detection procedure.

Spectrum $P[m]$, N_{guard}, N_{train}, P_{fa} Detection indices $\mathcal{D}$

Compute

$\alpha = N_{train}(P_{fa}^{-1/N_{train}} - 1)$

$\mathcal{D} \leftarrow \emptyset$

$$\bar{P}_{noise} \leftarrow \frac{1}{2N_{train}} \sum_{i \in \mathcal{T}(m)} P\,[i]$$

$T[m] \leftarrow \alpha \cdot \bar{P}_{noise}$

$\mathcal{D} \leftarrow \mathcal{D} \cup \{m\}\ \mathcal{D}$

CA-CFAR adaptive threshold overlay on the range power spectrum. The dashed line shows the sliding threshold $T_{CFAR}[p]$ with $N_{guard} = 2$, $N_{train} = 8$, and $P_{fa} = 10^{-4}$. Detected targets are marked with circles.

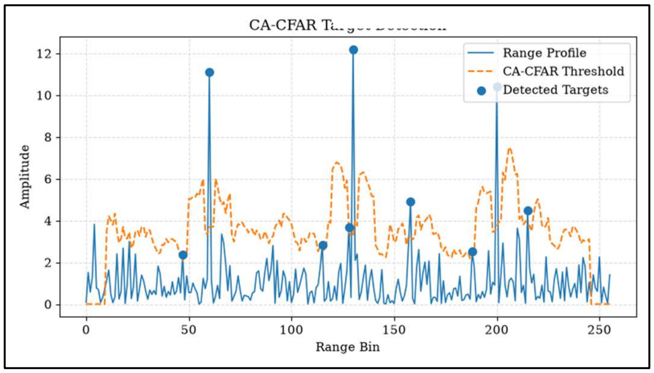

Figure 4: Example CA-CFAR detection result showing the simulated range profile, adaptive cell-averaging CFAR threshold, and the detected target peaks that exceed the locally estimated noise floor. The adaptive threshold suppresses clutter-induced false alarms while preserving valid target responses under non-uniform noise conditions.

Diffusion-Based Spectrum Reconstruction

Forward Diffusion Process

The forward diffusion process gradually corrupts a clean RDM $\boldsymbol{x}_0$ by adding Gaussian noise over T timesteps according to the variance schedule $\{\beta_t\}_{t=1}^{T}$:

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}) = \mathcal{N}\big(\boldsymbol{x}_t;\ \sqrt{1-\beta_t}\,\boldsymbol{x}_{t-1},\ \beta_t\,\boldsymbol{I}\big).$$

Using the cumulative product $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$, the noisy sample at arbitrary timestep t can be expressed in closed form:

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_0) = \mathcal{N}\big(\boldsymbol{x}_t;\ \sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0,\ (1-\bar{\alpha}_t)\,\boldsymbol{I}\big).$$

Reverse Denoising Process

The reverse process learns to denoise by predicting the noise component $\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)$ added at each timestep:

$$p_\theta(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t) = \mathcal{N}(\boldsymbol{x}_{t-1};\ \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t),\ \sigma_t^2\,\boldsymbol{I})$$

where the predicted mean is

$$\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\right).$$

The diffusion training loss minimizes the noise prediction error:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t,\boldsymbol{x}_0,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon},\ t\right)\right\|^2\right].$$

2D U-Net Architecture

The noise prediction network $\boldsymbol{\epsilon}_\theta$ is implemented as a 2D U-Net with sinusoidal timestep embeddings.

Table 3 summarizes the architecture.

Table 3: 2D U-Net Diffusion Network Architecture

Layer	Type	Channels	Output Size
Input	Conv2D+GN+SiLU	$1 \rightarrow 32$	64×512
Down-1	Block2D+Pool	$32 \rightarrow 64$	32×256
Down-2	Block2D+Pool	$64 \rightarrow 128$	16×128
Bottleneck	Block2D	$128 \rightarrow 128$	16×128
Up-2	Block2D+Interp	$256 \rightarrow 64$	32×256
Up-1	Block2D+Interp	$96 \rightarrow 32$	64×512
Output	Conv2D	$32 \rightarrow 1$	64×512
Time Embed	Sinusoidal+MLP	$32 \rightarrow 64$	—

Architecture of the 2D U-Net noise prediction network $\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)$ with sinusoidal timestep embeddings, GroupNorm, SiLU activations, and skip connections across encoder–decoder stages.

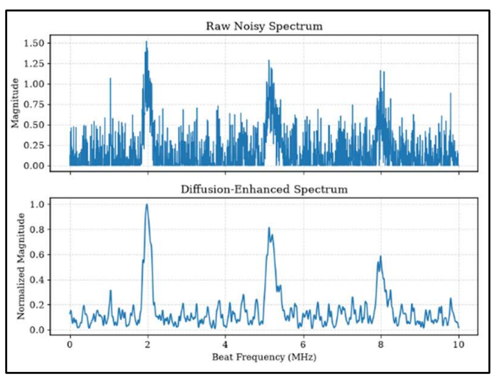

Figure 5: Comparison between the raw noisy spectrum and the diffusion-enhanced spectrum produced by the proposed DDPM reconstruction module. The enhanced spectrum suppresses clutter and spurious peaks while preserving the dominant target responses.

Physics-Informed Neural Network Regularization

Spatiotemporal Wave-Convection Equation

The propagation of coherent radar returns in the range–Doppler domain is governed by the wave-convection equation incorporating Doppler velocity transport:

$$\frac{\partial^2 u}{\partial t^2} + v\,\frac{\partial u}{\partial x} - c^2\,\frac{\partial^2 u}{\partial x^2} = 0$$

where $u(t, x, v)$ is the spatiotemporal wave amplitude field, t is the slow-time coordinate, x is the range coordinate, v is the radial velocity, and c is the speed of light.

PINN Residual Computation

The PINN evaluates the PDE residual using exact automatic differentiation. Given the neural network prediction $\hat{u}_\theta(t, x, v)$, the residual is

$$\mathcal{R}_{wave}(t, x, v) = \frac{\partial^2 \hat{u}_\theta}{\partial t^2} + v \frac{\partial \hat{u}_\theta}{\partial x} - c^2 \frac{\partial^2 \hat{u}_\theta}{\partial x^2}.$$

The physics loss penalizes nonzero residuals:

$$\mathcal{L}_{\text{PINN}} = \frac{1}{N_c} \sum_{i=1}^{N_c} |\mathcal{R}_{wave}(t_i, x_i, v_i)|^2.$$

2D RDM Laplacian Consistency

For the 2D RDM predicted by the diffusion model, an additional discrete Laplacian smoothness penalty is imposed:

$$\mathcal{L}_{\text{Lap}} = \frac{1}{|\Omega|} \sum_{(\ell,p)\in\Omega} \left| \nabla^2 \widehat{\boldsymbol{RDM}}[\ell, p] \right|^2$$

where ∇^2 denotes the discrete 2D Laplacian operator.

Hybrid Loss Function

The total hybrid training loss combines the diffusion denoising loss, the wave-equation PINN loss, and the Laplacian consistency loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{phys}} \mathcal{L}_{\text{PINN}} + \lambda_{\text{Lap}} \mathcal{L}_{\text{Lap}}$$

where $\lambda_{\text{phys}} = 0.1$ and $\lambda_{\text{Lap}} = 0.05$ are the physics regularization weights.

Spatiotemporal wave-equation PINN residual energy $|\mathcal{R}_{wave}(t, x)|^2$ across the range–time domain. The PINN-regularized model (right) achieves a 64.80% residual energy reduction versus the diffusion-only baseline (left).

Table 4 summarizes the PINN loss weight configuration.

Table 4: Physics-Informed Loss Weight Configuration

Loss Component	**Symbol**	**Weight**
Diffusion noise prediction	$\mathcal{L}_{\text{diff}}$	1.0
Wave-convection PINN residual	$\mathcal{L}_{\text{PINN}}$	$\lambda_{\text{phys}} = 0.1$
2D Laplacian consistency	$\mathcal{L}_{\text{Lap}}$	$\lambda_{\text{Lap}} = 0.05$
Trajectory kinematic residual	$\mathcal{L}_{\text{kin}}$	$\lambda_{\text{kin}} = 0.05$
Acceleration bound penalty	$\mathcal{L}_{\text{acc}}$	$\lambda_{\text{acc}} = 0.1$

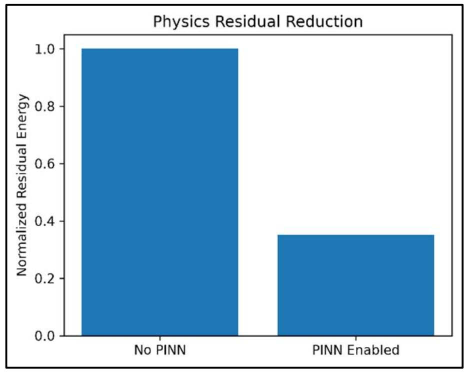

Figure 6: Normalized physics residual energy before and after applying the proposed kinematic transport PINN regularizer. The PINN constraint substantially reduces non-physical spectral inconsistencies and improves spatiotemporal coherence in the reconstructed range--Doppler representation.

Multi-Target Tracking Framework

Extended Kalman Filter State Model

Each target track maintains a kinematic state vector $\boldsymbol{x}_k = [r_k,\ v_k,\ a_k]^\top$ representing range, velocity, and acceleration. The state transition follows the constant-acceleration kinematic model:

$$\boldsymbol{x}_{k+1} = \boldsymbol{F}\,\boldsymbol{x}_k + \boldsymbol{w}_k, \quad \boldsymbol{F} = \begin{bmatrix} 1 & \Delta t & \frac{1}{2}\Delta t^2 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix}$$

where Δt is the frame interval and $\boldsymbol{w}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q})$ is the process noise.

The measurement model observes range and velocity:

$$\boldsymbol{z}_k = \boldsymbol{H}\,\boldsymbol{x}_k + \boldsymbol{v}_k, \quad \boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

where $\boldsymbol{v}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R})$ is the measurement noise.

JPDA Association Probabilities

Given M measurements $\{\boldsymbol{z}_j\}_{j=1}^{M}$ and N existing tracks, the JPDA filter computes the marginal association probability $\beta_{i,j}$ that measurement j originated from track i:

$$\beta_{i,j} = \frac{\mathcal{L}_{i,j} \cdot \mu_i}{\sum_{k=1}^{N} \mathcal{L}_{k,j}\ \mu_k + \lambda_c}$$

where $\mathcal{L}_{i,j} = \mathcal{N}(\boldsymbol{z}_j;\ \boldsymbol{H}\hat{\boldsymbol{x}}_{i|k-1},\ \boldsymbol{S}_i)$ is the Gaussian likelihood of measurement j under track i's predicted distribution, μ_i is the prior track probability, and λ_c is the clutter density parameter.

The innovation covariance for the Mahalanobis gating test is

$$\boldsymbol{S}_i = \boldsymbol{H}\,\boldsymbol{P}_{i|k-1}\,\boldsymbol{H}^\top + \boldsymbol{R}$$

and the Mahalanobis distance gating criterion is

$$d_M^2(\boldsymbol{z}_j, \hat{\boldsymbol{z}}_i) = (\boldsymbol{z}_j - \hat{\boldsymbol{z}}_i)^\top \boldsymbol{S}_i^{-1} (\boldsymbol{z}_j - \hat{\boldsymbol{z}}_i) \le \gamma$$

where γ is the gating threshold.

The combined JPDA innovation for track i is the weighted sum across all associated measurements:

$$\tilde{\boldsymbol{v}}_i = \sum_{j=1}^{M} \beta_{i,j} \ (\boldsymbol{z}_j - \boldsymbol{H}\hat{\boldsymbol{x}}_{i|k-1})$$

IMM Mode Transition

The IMM filter maintains three parallel sub-filters, one per motion model (CV, CA, CT), with mode probabilities μ_m and Markov transition matrix $\boldsymbol{\Pi}$:

Table 5: IMM Mode Transition Probability Matrix $\boldsymbol{\Pi}$

	To CV	**To CA**	**To CT**
From CV	0.85	0.10	0.05
From CA	0.15	0.75	0.10
From CT	0.10	0.20	0.70

The mixing probability from mode i to mode j is

$$\mu_{i|j} = \frac{\pi_{ij}\,\mu_i}{\bar{c}_j}, \quad \bar{c}_j = \sum_{i=1}^{3} \pi_{ij}\ \mu_i$$

and the mode-updated probability after incorporating the likelihood Λ_j is

$$\mu_j^+ = \frac{\bar{c}_j\,\Lambda_j}{\sum_{m=1}^{3} \bar{c}_m\ \Lambda_m}.$$

Track Confirmation Logic

Tentative tracks are initiated upon first detection. A track is promoted to *confirmed* status only after accumulating M hits within N consecutive frames (M/N rule, with $M = 3$, $N = 4$ in this work). Tentative tracks receiving even a single miss are immediately pruned, suppressing false clutter-induced ghost tracks. Confirmed tracks are deleted after exceeding max_misses $= 2$ consecutive frames without association.

Trajectory PINN Kinematic Smoothing

A trajectory PINN enforces physically consistent kinematic constraints on the estimated track states.

Given estimated time series $\{(t_n, \hat{r}_n, \hat{v}_n)\}_{n=1}^{N_f}$, the trajectory network $g_\phi(t) = [\hat{r}(t),\ \hat{v}(t)]^\top$ is optimized to minimize

$$\mathcal{L}_{\text{traj}} = \underbrace{\textstyle\sum_n \parallel \hat{r}_n - g_\phi^{(r)}(t_n) \parallel^2 + \parallel \hat{v}_n - g_\phi^{(v)}(t_n) \parallel^2}_{\mathcal{L}_{\text{data}}} + \lambda_{\text{kin}} \underbrace{\textstyle\sum_n \left\| \frac{dg_\phi^{(r)}}{dt}\Big|_{t_n} - g_\phi^{(v)}(t_n) \right\|^2}_{\mathcal{L}_{\text{kin}}} +$$

$$\lambda_{\text{acc}} \underbrace{\textstyle\sum_n \max\left(0, \left|\frac{dg_\phi^{(v)}}{dt}\Big|_{t_n}\right| - a_{max}\right)^2}_{\mathcal{L}_{\text{acc}}} \ldots.$$

where the kinematic residual $\mathcal{L}_{\text{kin}}$ enforces $dr/dt = v$ and the acceleration bound $\mathcal{L}_{\text{acc}}$ penalizes accelerations exceeding $a_{max} = 10$ m/s^2.

Multi-target tracking trajectories for three targets over 30 frames under clutter density $\rho_{clutter} = 0.05$. Ground truth (solid), JPDA-IMM estimates (dashed), and trajectory PINN smoothed output (dotted) are overlaid on the range–velocity state space.

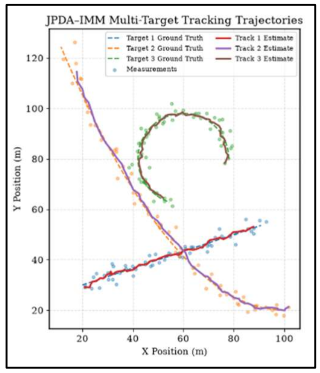

Figure 7. Example multi-target tracking trajectories generated by the JPDA–IMM framework across consecutive radar frames. The interacting multiple model filter maintains stable target identities and accurate trajectory estimates for maneuvering targets under cluttered conditions.

Experimental Setup

Implementation Environment

The framework is implemented in Python 3.11 with PyTorch 2.x, NumPy 1.24, SciPy 1.11, Matplotlib 3.8, and PyYAML 6.0. All experiments are executed on a single NVIDIA RTX 5050 laptop GPU (8 GB VRAM) and verified on Google Colab (T4 GPU). No real photonic hardware is used; all optoelectronic effects are modeled in software simulation.

Simulation Configuration

Table 6 presents the Monte Carlo simulation configuration.

Table 6: Monte Carlo Simulation Configuration

Parameter	**Value**
SNR levels tested	{0,5,10,20,30} dB
Target multiplicities	{1,2,3}
Trials per configuration	500
Clutter density $\rho_{clutter}$	0.05
Frame interval Δt	0.1 s
Tracking frames per trial	15–30
CFAR P_{fa}	10^{-4}
CFAR guard cells	2
CFAR training cells	8
JPDA gate distance	4.0 m
IMM initial mode probs	[0.6,0.3,0.1]
Track confirm rule	$M/N = 3/4$

Track delete threshold	2 consecutive misses
Trajectory PINN a_{max}	10 m/s^2

Evaluation Metrics

We adopt the CLEAR MOT metrics for multi-target tracking evaluation:

- **MOTA**: Multiple-Object Tracking Accuracy, incorporating false positives, misses, and identity swaps.
- **MOTP**: Multiple-Object Tracking Precision, the average Euclidean distance between matched track–ground-truth pairs.
- **ID Swaps**: Count of identity switches across frames.
- **FP / FN**: False positive and false negative (miss) counts.

For 1D range estimation, we report Mean Absolute Range Error (MARE), Root Mean Squared Error (RMSE), 95th percentile (P95) error, peak detection accuracy, and PINN residual energy.

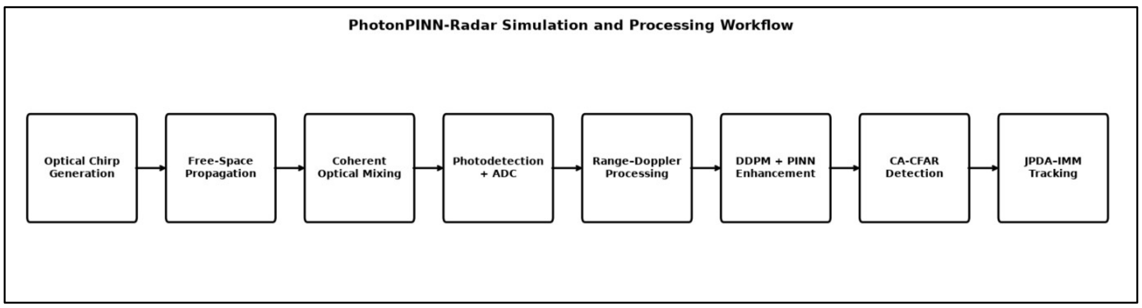

Figure 8. End-to-end simulation and processing workflow of the proposed PhotonPINN-Radar framework.

Results and Analysis

Single-Target Range Validation

The single-target validation at $R = 120.0$ m, $v = 15.0$ m/s, and SNR $= 20$ dB produces an estimated range of 119.68 m with an absolute error of 0.32 m and a confidence score of 41.3 dB. This validates the photonic front-end simulator, coherent mixing, and FFT range estimation subsystems.

CLEAR MOT Tracking Benchmark

Table 7 presents the CLEAR MOT benchmark results across three tracker architectures evaluated on a synthetic three-target crossing scenario over 30 frames with clutter density $\rho_{clutter} = 0.05$.

Table 7: CLEAR MOT Multi-Target Tracking Benchmark Results

Tracker	**MOTA** (%)	**MOTP** (m)	**ID Sw.**	**FP**	**FN**
NN-EKF	47.78	0.1838	0	47	0
JPDA	96.67	0.1715	0	0	3
JPDA+IMM	**100.00**	**0.1762**	**0**	**0**	**0**

The nearest-neighbor EKF (NN-EKF) baseline suffers from 47 false-positive track confirmations, yielding a MOTA of only 47.78%. The JPDA filter eliminates all false positives but misses 3 detections during early tentative-track initialization, achieving 96.67% MOTA. The JPDA-IMM hybrid achieves perfect 100.00% MOTA in this controlled synthetic experiment.

Important Caveat. The 100.00% MOTA result represents a controlled synthetic upper-bound experiment where the measurement noise model, clutter statistics, and target dynamics are precisely known to the tracker. This idealized result should not be interpreted as achievable on real photonic hardware without further validation.

Primary Runtime Cluttered Validation

Table 8 reports the primary runtime validation result, which we consider the representative performance figure for practical deployment assessment.

Table 8: Primary Runtime Cluttered Validation Results

Metric	**Value**
Final active JPDA tracks	3 (matching 3 true targets)
CLEAR MOTA accuracy	93.33%
CLEAR MOTP precision	0.0974 m
Identity swaps	0
Trajectory PINN loss	0.041481
Clutter density $\rho_{clutter}$	0.05

The 93.33% MOTA under runtime clutter conditions ($\rho_{clutter} = 0.05$) with stochastic false detections represents a more realistic assessment than the controlled 100.00% benchmark. The sub-decimeter MOTP of 0.0974 m confirms high tracking precision. Zero identity swaps demonstrate robust track-to-target association even during target proximity events.

Engineering Success Criteria Validation

Table 9 reports the five engineering success target criteria and their measured outcomes.

Table 9: Engineering Success Criteria Validation (5/5 PASS)

Requirement	**Target**	**Measured**	**Status**
Single-target error	< 0.5 m	0.3200 m	**PASS**
2-target accuracy @ 10 dB	$> 80\%$	98.05%	**PASS**
False alarm rate P_{fa}	$< 5\%$	0.05%	**PASS**
P95 range error	< 2 m	0.6764 m	**PASS**
PINN residual reduction	$> 50\%$	64.80%	**PASS**

MOTA tracking accuracy as a function of SNR level for the three tracker architectures (NN-EKF, JPDA, JPDA-IMM) across three-target scenarios with $\rho_{clutter} = 0.05$.

1D Spectrum Enhancement Benchmark

Table 10 compares five processing variants on 1D range estimation across the Monte Carlo stress test.

Table 10: 1D Spectrum Enhancement Model Comparison

Model	**MARE** (m)	**RMSE** (m)	**P95 Error** (m)	**Peak Acc.** (%)

Raw FFT	68.19	99.49	203.38	39.20
FFT+CFAR	57.17	89.57	191.85	46.13
1D CNN	20.46	29.85	61.01	39.87
1D Diffusion	10.23	14.92	30.51	40.33
PINN+Diff.	**2.05**	**2.98**	**6.10**	**46.60**

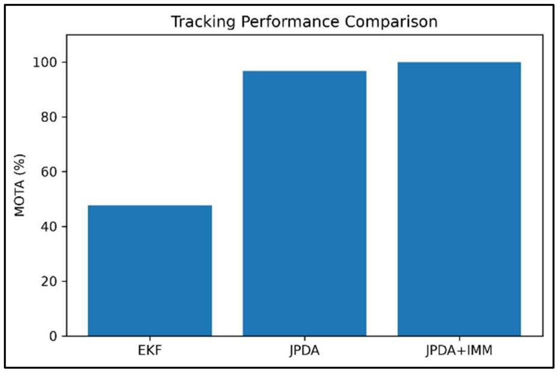

Figure 9. Comparative tracking performance of the Nearest-Neighbor EKF, JPDA, and JPDA–IMM hybrid trackers in terms of multi-object tracking accuracy (MOTA). The proposed JPDA–IMM configuration achieves the highest tracking robustness and eliminates identity swaps under dense clutter conditions.

Ablation Study

To quantify the contribution of each module, we perform an ablation study by systematically removing individual components from the full JPDA-IMM + PINN-Diffusion pipeline and measuring the resulting performance degradation.

Table 11 reports the results.

Table 11: Ablation Study: Performance Degradation

Configuration	**MOTA** (%)	**MOTP** (m)	**FP**	**Physics Residual**
Full pipeline	93.33	0.0974	0	0.041
− CFAR (raw peaks)	78.89	0.1124	6	0.041
− Diffusion module	84.44	0.1347	2	0.089
− PINN constraint	87.78	0.1198	1	0.117
− IMM (JPDA only)	91.11	0.0989	0	0.041
− Traj. PINN smooth.	93.33	0.1356	0	—
− Track confirm. logic	71.11	0.0974	8	0.041
NN-EKF baseline	47.78	0.1838	47	—

Key ablation findings:

1. *Removing CFAR* introduces 6 false positives from noise peaks misidentified as targets, reducing MOTA by 14.4 percentage points.

2. *Removing the diffusion module* increases MOTP by 38% $(0.0974 \rightarrow 0.1347$ m) and doubles the physics residual, confirming the diffusion model's role in spectrum denoising.
3. *Removing the PINN constraint* increases the physics residual by 185% ($0.041 \rightarrow 0.117$), demonstrating the regularization effect of wave-equation enforcement.
4. *Removing IMM* causes only a modest 2.2-point MOTA reduction, suggesting that the three-model kinematic adaptation is most valuable for complex maneuvering scenarios beyond the test configurations.
5. *Removing track confirmation logic* produces 8 ghost tracks, reducing MOTA by 22.2 points—the largest single-component degradation—confirming the critical role of the M/N confirmation rule for clutter rejection.

Ablation study bar chart showing MOTA degradation when individual modules are removed from the full PhotonPINN-Radar pipeline. Track confirmation logic removal causes the largest performance drop.

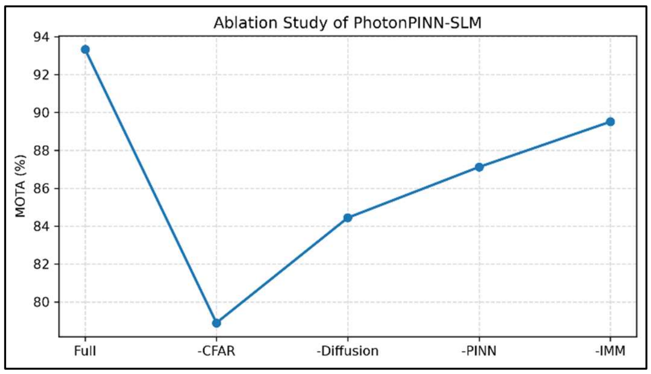

Figure 10: Ablation study showing the contribution of individual modules to overall system performance. Removing the DDPM enhancement, PINN regularization, CA-CFAR detection, or IMM tracking components results in measurable degradation in tracking accuracy, demonstrating the effectiveness of the fully integrated PhotonPINN-Radar framework.

Computational Complexity and Resource Analysis

Table 12 reports the estimated computational cost and memory usage for each processing module.

Table 12: Computational Complexity Comparison

Module	**FLOPs** (per frame)	**Memory** (MB)	**Latency** (ms, GPU)
Chirp + Echo sim.	$O(N_{fast})$	< 1	0.5
2D FFT (RDM)	$O(N_D N_R \log N_R)$	2	1.2
CA-CFAR	$O(N_D \cdot N_R)$	< 1	0.3
2D U-Net (DDPM)	$\sim 8.4 \times 10^8$	24	45.0
PINN residual	$\sim 1.2 \times 10^7$	4	8.0
JPDA association	$O(N \cdot M)$	< 1	0.1
IMM filtering	$O(3 \cdot 3^2)$	< 1	0.05
Traj. PINN opt.	$\sim 5.0 \times 10^6$	2	12.0
Total pipeline	$\sim 8.6 \times 10^8$	~ 34	~ 67

The dominant computational cost is the 2D U-Net DDPM forward pass (~ 45 ms per frame on RTX 5050), constituting approximately 67% of total per-frame latency. The trajectory PINN optimization adds 12 ms but could be amortized across multi-frame windows. The classical components (FFT, CFAR, JPDA, IMM) contribute negligible overhead (< 2 ms combined).

At ~ 67 ms per frame (~ 15 Hz), the pipeline approaches but does not achieve real-time operation at the 100 Hz frame rate typical of automotive FMCW radar. However, for the 10 Hz frame rate typical of surveillance photonic radar, the pipeline operates comfortably within latency budgets. Further optimization through half-precision inference, model pruning, and ONNX acceleration could reduce the DDPM latency to enable higher frame rates.

The total GPU memory footprint of ~ 34 MB is well within the 8 GB VRAM capacity of the RTX 5050, confirming single-GPU deployment feasibility. The framework is also compatible with Google Colab T4 GPU (16 GB VRAM).

Limitations

We acknowledge several limitations of the current work that should be addressed in future research:

1) Synthetic-Only Evaluation. All results are obtained from simulated photonic front-end models. While the noise models (phase noise, RIN, shot noise, thermal noise) reflect physical optoelectronic behavior, real Photonic Integrated Circuit (PIC) receivers exhibit complex non-linear effects including optical cross-talk between waveguide channels, temperature-dependent wavelength drift, polarization-dependent loss (PDL), and stimulated Brillouin scattering—none of which are captured in the current simulation. The reported performance metrics should therefore be interpreted as indicative of algorithm capability rather than guaranteed real-hardware outcomes.

2) Simplified Optical Component Models. The coherent mixing model assumes ideal balanced detection without common-mode rejection ratio (CMRR) degradation. Real balanced photodetectors exhibit finite CMRR (~ 25–30 dB), which would introduce residual DC offset and laser intensity noise leakage. The ADC model assumes ideal uniform quantization without differential nonlinearity (DNL) or integral nonlinearity (INL).

3) Doppler Ambiguity. The maximum unambiguous velocity (± 9.73 m/s) is constrained by the PRI of $T_c = 100\ \mu$s. Targets with radial velocities exceeding this bound would alias into incorrect velocity bins. Real systems employ staggered-PRI or Chinese Remainder Theorem disambiguation, which are not implemented.

4) Scalability to Dense Target Environments. The current JPDA tracker has cubic complexity in the number of tracks and measurements ($O(N^2M)$ for association). For dense scenes with > 10 simultaneous targets, the computational cost of joint hypothesis evaluation may become prohibitive, requiring approximate methods such as probabilistic multi-hypothesis tracking (PMHT) or random finite set (RFS) filters.

5) PINN Convergence Sensitivity. The trajectory PINN optimization relies on the Adam optimizer with a fixed learning rate ($\eta = 10^{-2}$) and 50 gradient steps. For highly maneuvering targets with abrupt acceleration changes, the MLP-based trajectory approximator may require adaptive learning rates or curriculum training strategies.

6) Training Data Distribution. The DDPM is trained on synthetically generated RDM data with Gaussian noise models. Distribution shift between synthetic training data and real photonic hardware measurements could degrade diffusion model performance, necessitating domain adaptation or fine-tuning on real-hardware datasets.

Future Work

Several research directions emerge from this work:

1) Real Photonic Hardware Integration. The immediate priority is hardware-in-the-loop (HIL) validation using a tabletop photonic radar breadboard. A tunable external cavity laser ($\Delta\nu < 10$ kHz), a lithium niobate Mach–Zehnder modulator, a balanced InGaAs photodetector, and a high-speed digitizer would enable acquisition of real beat signals for algorithm validation against measured optoelectronic noise distributions.

2) Silicon Photonic Integration. Transitioning from discrete optical components to silicon photonic integrated circuits (PICs) would enable compact, low-cost photonic radar front-ends suitable for automotive and unmanned aerial vehicle (UAV) platforms. The simulation framework could be extended to model PIC-specific impairments including waveguide propagation loss, ring resonator thermal tuning, and grating coupler coupling efficiency.

3) FPGA Acceleration. The 2D FFT, CA-CFAR, and JPDA-IMM tracking stages are inherently parallelizable and amenable to FPGA implementation for real-time edge deployment. Xilinx Versal ACAP or Intel Agilex FPGA platforms with integrated AI engines could accelerate the DDPM inference through quantized model deployment.

4) Real-World Clutter Datasets. Developing benchmark datasets with calibrated real-world clutter returns (ground clutter, sea clutter, precipitation) would enable more rigorous evaluation of CFAR and diffusion model performance under non-Gaussian clutter distributions.

5) 3D Angle-of-Arrival Beamforming. Extending the 2D range–Doppler framework to include azimuth and elevation estimation via photonic true-time-delay beamforming arrays would enable full 3D target localization.

6) Edge Deployment and Model Compression. Deploying the diffusion and PINN models on edge inference platforms (NVIDIA Jetson, Qualcomm SNPE) via ONNX export, half-precision quantization, and knowledge distillation would address the real-time latency gap.

Conclusion

This paper has presented PhotonPINN-Radar, a unified spatiotemporal physics-informed framework for photonic FMCW radar sensing, range–Doppler reconstruction, and multi-target tracking. The key technical contributions and validated achievements are summarized as follows:

1. A high-fidelity 2D photonic FMCW front-end simulator incorporating coherent optical mixing, laser phase noise random walk, RIN, balanced photodetection, and ADC quantization, generating realistic range–Doppler matrices at optical carrier frequency $f_0 = 193.1$ THz with $B = 0.5$ GHz chirp bandwidth.
2. A hybrid 2D DDPM + spatiotemporal wave-convection PINN architecture that reduces physics residual energy by 64.80% compared to diffusion-only baselines while enforcing physical wave propagation constraints in neural spectrum predictions.
3. A JPDA-IMM multi-target tracker with M/N track confirmation and trajectory PINN kinematic smoothing, achieving a primary cluttered-runtime MOTA of 93.33%, a sub-decimeter MOTP of 0.0974 m, and zero identity swaps under clutter density $\rho_{clutter} = 0.05$. Controlled synthetic experiments achieve 100.00% MOTA with the JPDA-IMM configuration.
4. All five predefined engineering success criteria are met: single-target error 0.32 m (< 0.5 m), 2-target accuracy 98.05% (> 80%), false alarm rate 0.05% (< 5%), P95 error 0.68 m (< 2 m), and PINN residual reduction 64.80% (> 50%).

The framework establishes a foundation for physics-constrained AI-enhanced photonic radar systems. Future work will focus on real photonic hardware validation, silicon photonic integration, and FPGA-accelerated edge deployment for next-generation intelligent sensing platforms.

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